Morphic Studio AI Intelligence & Knowledge Architecture

Document 73 — Knowledge Graph, Semantic Intelligence & Canon Reasoning Specification

Version: 1.0

Status: Approved

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1. Purpose

This document defines the Knowledge Graph, Semantic Intelligence, and Canon Reasoning Framework for Morphic Studio.

Its purpose is to organize creative knowledge as an interconnected semantic network that enables AI agents to understand relationships, infer meaning, validate consistency, support intelligent retrieval, and reason across complex fictional universes and production workflows.

The Knowledge Graph becomes the platform's shared understanding of every creative project.

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2. Vision

Creative information should not exist as isolated documents.

Instead, every meaningful entity becomes part of a living semantic network.

Examples include:

- Characters
- Stories
- Chapters
- Scenes
- Worlds
- Locations
- Objects
- Organizations
- Timelines

- Events
- Assets
- Creative decisions

Relationships are treated as first-class knowledge.

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3. Design Principles

The Knowledge Graph follows these principles:

- Relationships are as important as entities.
- Knowledge should remain explainable.
- Canon consistency is continuously evaluated.
- Graph evolution preserves history.
- Reasoning should remain transparent.
- Retrieval prioritizes semantic relevance.
- Human authors remain authoritative.

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4. Core Entity Types

The graph supports entities including:

Creative Entities

- Story
- Chapter
- Scene
- Character
- Dialogue
- Plot Arc

World Entities

- Planet
- Region
- City
- Building
- Location
- Civilization
- Faction
- Culture

Production Entities

- Asset
- Illustration
- Animation
- Audio
- Video
- Workflow
- Publication

Operational Entities

- User
- Team
- Organization
- AI Agent
- Workflow Stage
- Review

The schema remains extensible.

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5. Relationship Model

Relationships may include:

- Belongs To
- Located In
- Created By
- Depends On
- Uses
- Owns
- Knows
- Opposes
- Supports
- Mentors
- Descends From
- Appears In
- Participated In
- Influences
- References
- Conflicts With

Relationships may include:

- Direction
- Strength

- Validity period
- Confidence
- Canon status
- Provenance

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6. Timeline Graph

Events become graph nodes.

Relationships capture:

- Before
- After
- Concurrent
- Causes
- Prevents
- Modifies
- References

Timeline reasoning supports chronological consistency.

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7. Character Network

Character relationships include:

- Family
- Friendship
- Rivalry
- Leadership
- Romance
- Alliance
- Employment
- Mentorship
- Historical interactions

Relationship history is preserved across story evolution.

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8. Canon Graph

Canon reasoning tracks:

- Established facts
- Canon hierarchy
- Official revisions
- Retcons
- Contradictions
- Alternative timelines
- Multiverse branches
- Version lineage

Canon remains queryable and explainable.

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9. Semantic Search

Search leverages graph intelligence through:

- Entity discovery
- Relationship traversal
- Intent-aware search
- Similarity ranking
- Context expansion
- Canon-aware retrieval
- Cross-project linking

Semantic search complements traditional text search.

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10. Graph Reasoning

AI agents perform reasoning including:

- Relationship inference
- Missing connection detection
- Contradiction analysis
- Timeline validation
- Canon verification
- Entity clustering
- Dependency analysis
- Knowledge completion

Reasoning outputs remain explainable.

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11. Graph Evolution

Knowledge evolves through:

Creation

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Validation

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Relationship Expansion

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Canonical Approval

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Historical Preservation

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Future Revision

Graph history remains permanently traceable.

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12. Cross-Project Knowledge

Organizations may reuse knowledge including:

- Species
- Technology
- Templates
- Style guides
- Production standards
- Shared assets
- Creative methodologies

Cross-project reuse respects organizational permissions.

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13. Explainable Intelligence

Every semantic conclusion may expose:

- Traversed entities
- Relationship paths
- Confidence scores
- Canon references
- Supporting evidence
- Alternative interpretations

Users can inspect reasoning before accepting AI recommendations.

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14. Governance

Knowledge governance defines:

- Entity ownership
- Relationship validation
- Canon approval
- Version control
- Access permissions
- Retention policies
- Archival procedures

Graph integrity remains continuously protected.

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15. Monitoring

Operational metrics include:

- Graph growth
- Relationship density
- Retrieval accuracy
- Canon conflict frequency
- Traversal performance
- Semantic search quality
- Knowledge freshness

Monitoring informs optimization.

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16. Integration

The Knowledge Graph integrates with:

- Multi-Agent Orchestrator
- Memory Framework
- Story Workspace
- Character Studio
- World Builder
- Story Bible
- Workflow Engine
- Asset Manager
- Publishing Center
- Analytics
- Administration

Every AI agent uses the same semantic foundation.

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17. Acceptance Criteria

The Knowledge Graph is successful if:

- AI understands relationships rather than isolated documents.
- Canon reasoning identifies inconsistencies accurately.
- Semantic search retrieves contextually relevant information.
- Timeline reasoning preserves chronological integrity.
- Graph evolution remains fully traceable.
- Cross-project knowledge sharing respects governance.
- Users can inspect and understand AI reasoning paths.

Conclusion

The Morphic Studio Knowledge Graph, Semantic Intelligence, and Canon Reasoning Framework transforms creative information into a living, interconnected knowledge ecosystem. Through semantic relationships, explainable graph reasoning, timeline intelligence, canon validation, and governed knowledge evolution, the platform enables AI collaborators to reason across complex creative universes with depth, consistency, and transparency while preserving creator authority and long-term project integrity.